

DEPARTMENT OF MATHEMATICS & STATISTICS

MTH 211: THEORY OF STATISTICS

Quiz 2 Solutions

Session: 2025-26

Semester: (II)

Question 1:

The joint likelihood function for a sample of size $n = 4$ from a $Poisson(\lambda)$ distribution is:

$$L(\lambda) = \prod_{i=1}^4 \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} = \frac{e^{-4\lambda} \lambda^{\sum x_i}}{\prod_{i=1}^4 x_i!}$$

Let $y = \sum_{i=1}^4 x_i$. For any $\lambda_2 > \lambda_1$, the ratio of the likelihood functions is:

$$\frac{L(\lambda_1)}{L(\lambda_0)} = \frac{e^{-4\lambda_1} \lambda_1^y}{e^{-4\lambda_0} \lambda_0^y} = e^{-4(\lambda_1 - \lambda_0)} \left(\frac{\lambda_1}{\lambda_0} \right)^y$$

Since $\lambda_1 > \lambda_0$, the NP lemma rejects the $MP(\alpha)$ test for $H_0 : \lambda = 1$ vs $H_A : \lambda = \lambda_1 (> 1)$ rejects H_0 for $Y = (\sum_{i=1}^4 X_i) > k$.

Hence, the non-randomized for an exact size- α MP test is given by:

(a) MP Test for $H_0 : \lambda = 1$ vs $H_A : \lambda = \lambda_1 (> 1)$):

By the Neyman-Pearson Lemma, the most powerful test of size $\alpha = 0.05$ is:

$$\phi(y) = \begin{cases} 1 & \text{if } y > k \\ \gamma & \text{if } y = k \\ 0 & \text{if } y < k \end{cases}$$

Under H_0 , $Y \sim \text{Poisson}(4)$. We seek $E_{H_0}[\phi(Y)] = P(Y > k) + \gamma P(Y = k) = 0.05$.

Given $P(Y \geq 8) = 0.0511$ and $P(Y \geq 9) = 0.0214$:

- $P(Y = 8) = 0.0511 - 0.0214 = 0.0297$.

- Since $P(Y \geq 8) > 0.05$, we set $k = 8$.

Solving for γ : $0.0214 + \gamma(0.0297) = 0.05 \implies \gamma = \frac{0.05 - 0.0214}{0.0297} \approx 0.963$.

Since, the above test does not depend on any particular choice of $\lambda_1 (> 1)$, this is a $UMP(\alpha)$ Test for $H_0 : \lambda = 1$ vs $H_A : \lambda > 1$.

(b) UMP Test for $H_0 : \lambda \leq 1$ vs $H_A : \lambda > 1$:

To show $\phi(y)$ is UMP for the composite null, consider the power function

$$\beta_\phi(\lambda) = (1 - \gamma) \sum_{t=k+1}^{\infty} f_\lambda(t) + \gamma \sum_{t=k}^{\infty} f_\lambda(t).$$

By connecting this with the incomplete Gamma function, we show $\beta_\phi(\lambda)$ is continuous and **increasing** in λ . Thus:

$$\beta_\phi(\lambda) \leq \beta_\phi(1) = 0.05 \quad \forall \lambda \leq 1$$

and hence, the test derived in (a) is the UMP test for the composite null.

Question 2:

The likelihood function for the Rayleigh distribution is:

$$L(\sigma) = \prod_{i=1}^n \frac{x_i}{\sigma^2} e^{-x_i^2/(2\sigma^2)} = \frac{\prod x_i}{\sigma^{2n}} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n x_i^2\right)$$

For $\sigma_2 > \sigma_1$, the ratio $\frac{L(\sigma_2)}{L(\sigma_1)}$ is proportional to $\exp\left(\left[\frac{1}{2\sigma_1^2} - \frac{1}{2\sigma_2^2}\right] \sum x_i^2\right)$.

Since $\sigma_2 > \sigma_1 \implies \frac{1}{2\sigma_1^2} > \frac{1}{2\sigma_2^2}$, the exponent coefficient is positive. Thus, the ratio is an increasing function of $T = \sum X_i^2$, establishing the MLR property in T .

The UMP test rejects H_0 if $T = \sum X_i^2 > c$. Under H_0 , $\frac{1}{\sigma_0^2} \sum X_i^2 \sim \chi_{2n}^2$.

To satisfy size α : $P_{H_0}(T > c) = \alpha \implies P\left(\frac{T}{\sigma_0^2} > \frac{c}{\sigma_0^2}\right) = \alpha$.

Critical Region: $T > \sigma_0^2 \cdot \chi_{2n,\alpha}^2$.

Question 3:

1. Hypotheses & Parameter Spaces:

$$\Omega = \{\mu \in \mathbb{R}\}, \quad \Omega_0 = \{\mu = \mu_0\}, \quad \Omega_A = \{\mu < \mu_0\}$$

2. Likelihood functions:

$$L(\mu) = \left(\frac{1}{2\pi\sigma^2}\right)^{n/2} \exp\left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right]$$

Under Ω_0 , $L(\mu) = L(\mu_0)$.

Under $\Omega_0 \cup \Omega_A$, $\sup_{\mu \leq \mu_0} L(\mu) = \begin{cases} L(\mu_0) & \text{if } \bar{x} > \mu_0 \\ L(\bar{x}) & \text{if } \bar{x} \leq \mu_0 \end{cases}$

3. Likelihood Ratio Statistic (Λ):

$$\lambda(x) = \frac{L(\mu_0)}{\sup_{\mu \leq \mu_0} L(\mu)} = \begin{cases} 1 & \text{if } \bar{x} > \mu_0 \\ e^{-\frac{n}{2\sigma^2}(\bar{x} - \mu_0)^2} & \text{if } \bar{x} \leq \mu_0 \end{cases}$$

4. Decision Rule: The LR test rejects H_0 if $\lambda(x) < k$. Since the LR test always accepts H_0 for $\bar{x} > \mu_0$, the rejection criterion is equivalent to:

$$e^{-\frac{n}{2\sigma^2}(\bar{x} - \mu_0)^2} < k \iff (\bar{x} - \mu_0)^2 > k^* \iff \bar{x} < c$$

NOT to consider the case $\bar{x} > c^*$ though it is also implied by

$$(\bar{x} - \mu_0)^2 > k^*$$

5. Size Condition: Under H_0 , $P(\bar{X} < c) = \alpha \implies P\left(\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} < \frac{c - \mu_0}{\sigma/\sqrt{n}}\right) = \alpha$.

$$\frac{c - \mu_0}{\sigma/\sqrt{n}} = z_{1-\alpha} = -z_\alpha \implies c = \mu_0 - z_\alpha \sigma / \sqrt{n}$$